Blood Bank Management System

Software Requirements Specification

Part A Software Requirements Specification

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1.0	29 September 2026	Complete part a software requirements specification for Part 1 submission	Project team

Approval Record

Role	Name	Signature	Date
Faculty evaluator			
Team representative			

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1 Introduction

1.1 Purpose

This Software Requirements Specification defines the behaviour, interfaces, constraints and quality requirements of the Blood Bank Management System. It provides the baseline for design, implementation, testing and acceptance.

1.2 Scope

The system supports one or more blood-bank branches under one deployment. It maintains donor and donation records, applies configurable eligibility rules, tracks laboratory screening and component inventory, accepts hospital blood requests, records cross-matching and issuance, sends operational notifications and produces audit and management reports.

The first implementation may operate at one branch. The data model and access controls retain branch identifiers so that additional branches can be enabled without replacing the core design.

1.3 Intended Audience

The intended readers are the course evaluator, project team, student developers and student testers. Operational stakeholders include donors, blood-bank staff, hospital staff and system administrators.

1.4 Definitions and Abbreviations

Term	Meaning
API	Application Programming Interface
Blood unit	A uniquely identified whole-blood collection or derived component tracked by the system
Component	A blood product such as red cells, platelets or plasma
CSRF	Cross-Site Request Forgery
FR	Functional Requirement
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
RTM	Requirements Traceability Matrix
SRS	Software Requirements Specification
UTC	Coordinated Universal Time

1.5 References

The normative and project references are listed in the References section at the end of this document.

1.6 Document Conventions

The word shall indicates a mandatory and testable requirement. Functional requirements use FR identifiers, non-functional requirements use NFR identifiers and security requirements use SEC identifiers. Priorities indicate implementation importance for the academic prototype.

2 Overall Description

2.1 Product Perspective

The product is a role-based web application backed by a relational database. It centralizes records that may otherwise be kept in separate registers or spreadsheets. Browser clients communicate with the application over HTTPS. The application sends email through a configured provider and stores encrypted backups in restricted storage.

2.2 Product Functions

- Register donors and maintain consent, contact and medical-history information.
- Evaluate eligibility through configurable rules and active deferrals.
- Schedule donation appointments with branch and slot capacity controls.
- Record collection, component preparation, laboratory testing and unit state changes.
- Search inventory and manage hospital blood requests, reservations, cross-matches and issuance.
- Generate alerts, reports, backups and audit records.

2.3 User Classes

User class	Main responsibilities	Access level
Donor	Profile, eligibility, appointments and personal history	Own records only
Blood Bank Staff	Screening, donation, testing, inventory, requests and issue	Assigned branch operations
Hospital Staff	Availability search, request submission and request tracking	Hospital requests and restricted stock view
System Administrator	Accounts, roles, branches, configuration and reports	Administrative data without routine clinical editing

2.4 Operating Environment

- Server operating system: Linux.
- Application: Python and Django served by Gunicorn behind Nginx.
- Database: PostgreSQL through the Django object-relational mapper.
- Client: Supported desktop or mobile browser with JavaScript and cookies enabled.
- Deployment: Docker containers may be used, but containerization is not required for the academic prototype.

2.5 Design Constraints

- Server-side authorization is mandatory for protected operations.
- All persistent clinical and inventory records use database transactions and referential integrity.
- Eligibility, shelf-life and notification thresholds remain configurable because approved rules may change.
- The interface uses forms and tables suitable for a semester-scale implementation.

2.6 Assumptions and Dependencies

- Each hospital and staff account is verified by an administrator before activation.
- Laboratory results are entered by authorized staff; direct analyser integration is outside the first release.
- The deployment provides working email, database and backup services.
- A qualified authority approves operational eligibility and retention rules before deployment.

2.7 Items Outside Scope

- Payment processing and insurance claims.
- Direct control of laboratory analysers or storage hardware.
- Biometric or national identity integration.
- Automated clinical diagnosis or replacement of professional cross-match review.

3 External Interface Requirements

3.1 User Interfaces

Role-specific dashboards present only permitted actions. Forms display mandatory-field markers, inline validation and a summary of submission errors. Tables support paging, filtering and clear empty states. Destructive actions require confirmation. Status values use text labels and do not rely on colour alone.

3.2 Hardware Interfaces

No dedicated hardware interface is required for the first release. Staff use standard desktop or tablet input devices. Barcode support may be added later through keyboard-compatible scanners.

3.3 Software Interfaces

Interface	Purpose	Failure handling
PostgreSQL	Persistent transactional storage	Rollback transaction and log a database error
SMTP email provider	Appointments and operational notifications	Record failure and retry through a scheduled task
Backup storage	Encrypted database backup retention	Alert administrator and preserve prior successful backups

3.4 Communication Interfaces

Browser traffic uses HTTPS. API payloads use JSON encoded as UTF-8. Server-generated timestamps use ISO 8601 UTC format. Email transport uses the configured authenticated and encrypted provider connection.

4 System Requirements

4.1 Requirement Quality Rules

Every requirement has one identifier, one primary behaviour and one verification reference. Quantitative limits are included where acceptance depends on time, load, retention or viewport size. Configurable values are identified instead of being embedded in application code.

4.2 Functional Requirements

Table 1 Functional requirements

ID	Requirement	Priority	Verification
FR-01	The system shall register a donor after validating mandatory identity, contact, consent and medical history fields.	High	Inspection and TC-F01
FR-02	The system shall maintain one donor profile per registered email address and shall reject a duplicate active profile.	Medium	TC-F01
FR-03	The system shall evaluate donor eligibility using administrator-configured rules for age, weight, donation type, donation interval, deferrals and screening responses.	High	TC-F02
FR-04	The system shall display each failed eligibility rule and shall prevent appointment confirmation when any blocking rule fails.	High	TC-F02
FR-05	The system shall allow an eligible donor to create, reschedule or cancel an appointment for an available branch and time slot.	High	TC-F03
FR-06	The system shall prevent appointment capacity from exceeding the configured capacity of a time slot.	High	TC-F03
FR-07	The system shall display the authenticated donor's donation history, appointment history and eligibility date.	Medium	TC-F04
FR-08	The system shall allow authorized staff to record temporary or permanent donor deferrals with reason, start date, end date when applicable and approving staff member.	High	TC-F05
FR-09	The system shall assign a unique donation identifier and unique blood unit identifier to every recorded donation and derived component.	High	TC-F06
FR-10	The system shall record collection time, volume, blood group, component type, collector and source donor for each blood unit.	High	TC-F06
FR-11	The system shall store laboratory results for the configured mandatory transfusion-transmitted infection test panel for each unit.	High	TC-F07
FR-12	The system shall prevent a unit from entering the Available state until all mandatory test results are recorded as acceptable by authorized staff.	High	TC-F07
FR-13	The system shall place a unit in Quarantined state when any mandatory screening result is reactive, indeterminate or incomplete.	High	TC-F07
FR-14	The system shall manage unit states as Collected, Testing, Available, Reserved, Issued, Quarantined, Expired or Discarded and shall reject an invalid state transition.	High	TC-F08

ID	Requirement	Priority	Verification
FR-15	The system shall calculate expiry from collection time, component type and the configured shelf-life rule.	High	TC-F08
FR-16	The system shall mark an unissued unit Expired when its expiry time is reached and shall exclude it from available stock.	High	TC-F08
FR-17	The system shall allow authorized users to search available stock by blood group, component, branch and expiry date.	High	TC-F09
FR-18	The system shall notify designated staff when available stock falls below a configured threshold or when a unit enters the configured near-expiry period.	Medium	TC-F10
FR-19	The system shall allow hospital staff to create a request containing blood group, component, quantity, priority, required-by time and clinical reference.	High	TC-F11
FR-20	The system shall support Pending, Approved, Partially Fulfilled, Fulfilled, Rejected and Cancelled request states and shall record the actor and time of every transition.	High	TC-F11
FR-21	The system shall allow hospital staff to cancel only a Pending request and shall require blood bank staff to cancel a request after approval.	Medium	TC-F12
FR-22	The system shall allow authorized staff to reserve compatible Available units for an approved request without assigning one unit to more than one active request.	High	TC-F13
FR-23	The system shall record cross-match outcome before a reserved unit is issued.	High	TC-F13
FR-24	The system shall complete unit issuance and request status update in one database transaction.	High	TC-F13
FR-25	The system shall authenticate Donor, Blood Bank Staff, Hospital Staff and System Administrator accounts before protected functions are accessed.	High	TC-S01
FR-26	The system shall allow a system administrator to create, deactivate and assign one approved role to a user account.	High	TC-S02
FR-27	The system shall allow a system administrator to manage branches, time-slot capacities, eligibility rules, shelf-life rules and stock thresholds.	Medium	TC-F14
FR-28	The system shall generate inventory, expiry, donation, request and audit reports for a selected date range and branch.	Medium	TC-F15
FR-29	The system shall send appointment reminders and request-status notifications through the configured email service and shall record delivery success or failure.	Low	TC-F16
FR-30	The system shall write an audit event for authentication, deferral, screening, inventory, request, issuance, configuration and account-management changes.	High	TC-S03

4.3 Non-Functional Requirements

Table 2 Non-functional requirements

ID	Requirement	Quality area	Verification
NFR-01	For at least 95 percent of search and dashboard requests, the server shall return a response within 3 seconds with 100 concurrent authenticated users in the defined test environment.	Performance	TC-N01
NFR-02	The system shall complete a stock reservation transaction within 5 seconds under the same load and shall prevent duplicate reservation under concurrent requests.	Performance	TC-N02
NFR-03	The monthly service availability target shall be 99.0 percent, excluding planned maintenance announced at least 24 hours in advance.	Reliability	Operational review
NFR-04	A scheduled encrypted backup shall run every 24 hours, retain seven daily copies and four weekly copies, and support restoration within 4 hours during the acceptance test.	Recovery	TC-N03
NFR-05	The system shall preserve committed donation, screening, reservation and issuance data after an application restart.	Reliability	TC-N04
NFR-06	The interface shall support the latest two major versions of Chrome, Edge and Firefox and viewport widths from 360 to 1440 pixels without horizontal scrolling on standard forms.	Compatibility	TC-N05
NFR-07	A first-time staff user shall complete donor lookup and appointment check-in within 5 minutes using the supplied task instructions and without developer assistance.	Usability	TC-N06
NFR-08	The application shall separate presentation, business rules and data-access code into documented modules with automated unit tests for eligibility, unit state transitions and request state transitions.	Maintainability	Code review
NFR-09	The database schema shall support multiple branches while enforcing branch-level access restrictions for operational users.	Scalability	TC-S02
NFR-10	The application shall record dates and times in UTC and display them in the branch time zone configured by the administrator.	Data integrity	TC-N07
NFR-11	The system shall not expose medical-history or screening-result fields in availability search results or hospital-facing reports.	Privacy	TC-S04
NFR-12	The system shall support retention and export rules configured to match the approved institutional and regulatory record-retention policy.	Compliance	Inspection

5 Data Requirements

5.1 Core Data Entities

The core entities are User, Donor, Branch, Appointment, Deferral, Donation, BloodUnit, LabResult, BloodRequest, Reservation, CrossMatch, Issue, Notification and AuditEvent. Every clinical and inventory record has a stable identifier and creation timestamp.

5.2 Data Integrity Rules

- A blood unit identifier is unique and cannot be reused.
- A lab result references one valid blood unit and one configured test type.
- An issued unit references one issue transaction and cannot return to Available state.
- A reserved unit belongs to at most one active request.
- Deletion of clinical, inventory and audit records is disabled through normal application functions.

5.3 Retention and Privacy

Retention periods are configuration values approved by the institution. Export and archival processes preserve identifiers, timestamps and audit references. Availability searches expose stock counts and unit suitability fields but do not expose donor medical history or laboratory details to hospital accounts.

6 Security Objectives and Requirements

6.1 Security Objectives

1. Protect the confidentiality of donor, medical and request information.
2. Preserve the integrity of screening, inventory, cross-match and issuance records.
3. Permit each role to perform only authorized actions within its assigned scope.
4. Maintain accountable and reviewable records of security-sensitive and operational changes.
5. Support recovery from data loss without exposing backup contents.

6.2 Security Requirements

Table 3 Security requirements

ID	Requirement	Verification
SEC-01	The system shall enforce server-side authorization for every protected action according to the role-permission matrix.	TC-S02
SEC-02	The system shall store passwords using Django's approved adaptive password hashing mechanism and shall never store or log a plain-text password.	Code review
SEC-03	The system shall lock an account for 15 minutes after five consecutive failed login attempts within 10 minutes and shall record the event.	TC-S01
SEC-04	The system shall expire an authenticated session after 15 minutes of inactivity and shall invalidate the session after logout or account deactivation.	TC-S05
SEC-05	The production deployment shall redirect HTTP to HTTPS and shall transmit authenticated traffic only over TLS.	Configuration inspection
SEC-06	The system shall protect state-changing requests against cross-site request forgery and shall validate and encode user-controlled input before storage or display.	TC-S06
SEC-07	Audit records shall include actor ID, event type, UTC timestamp, affected record ID and outcome and shall not be editable through normal application roles.	TC-S03

ID	Requirement	Verification
SEC-08	Database backups shall be encrypted and accessible only to the backup service account and designated administrator.	TC-N03
SEC-09	The system shall display only the minimum donor and patient information required for the current role and task.	TC-S04
SEC-10	The system shall return a generic authentication failure message that does not reveal whether an email address is registered.	TC-S01

6.3 Role Permission Matrix

Table 4 Role permission matrix

Capability	Donor	Blood Bank Staff	Hospital Staff	Administrator
Own profile and appointments	Create and update	View for operations	No access	Account status only
Donor medical history	Own record	Authorized view and update	No access	No routine access
Laboratory results	No access	Create and view	No access	No routine access
Inventory	No access	Full branch operations	Restricted availability	Reports and configuration
Blood requests	No access	Approve and fulfil	Create and track own hospital	Reports only
Accounts and configuration	No access	No access	No access	Create and manage
Audit records	No access	Restricted read	No access	Read and export

7 Use Case Model

7.1 Actors

The primary actors are Donor, Blood Bank Staff, Hospital Staff and System Administrator. Email and scheduled-task services support system behaviour but are not human actors.

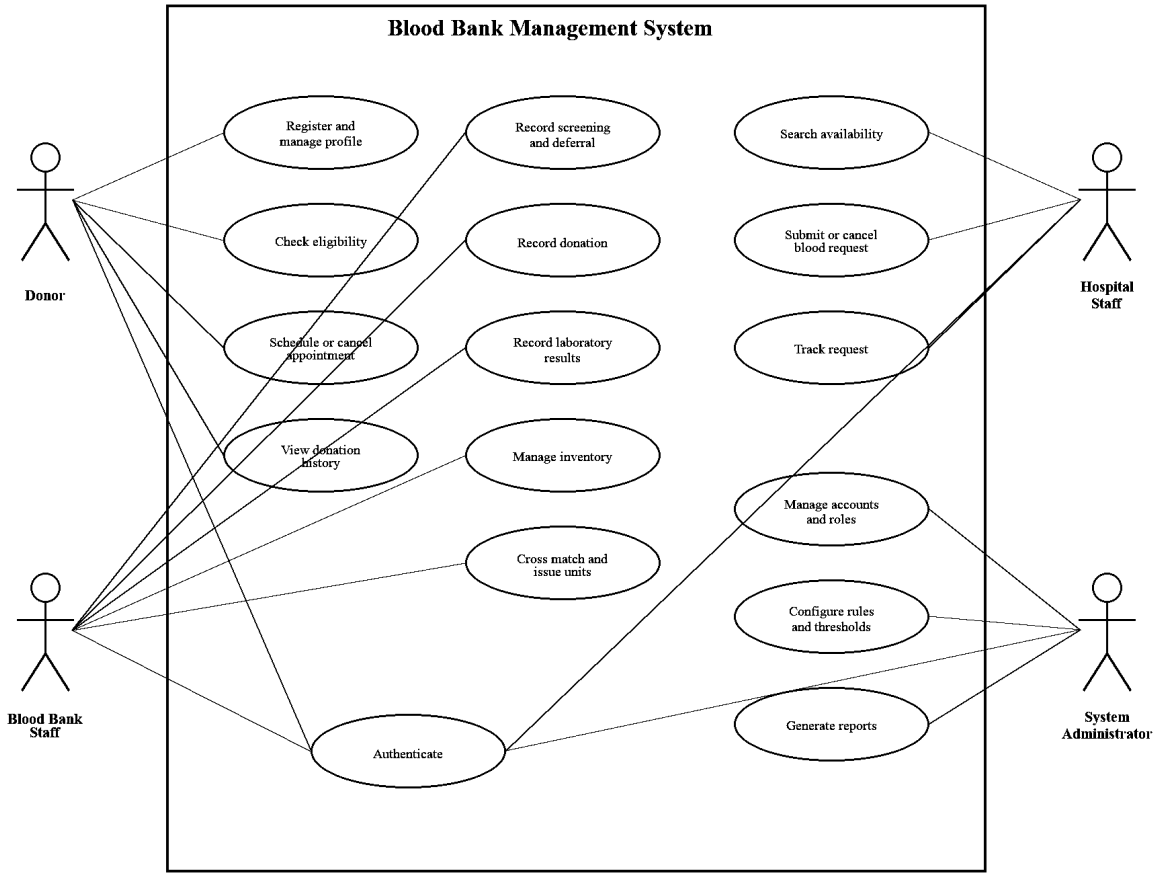


Figure 1 UML use case diagram

7.2 Detailed Use Cases

UC-01 Check Donor Eligibility

Field	Specification
Primary actor	Donor or Blood Bank Staff
Preconditions	An identified donor profile exists.
Trigger	An eligibility check is requested.
Main flow	The system loads current profile data, donation history, active deferrals and configured rules. It evaluates each rule and returns Eligible or Not Eligible with reasons.
Alternate and exception flow	Missing mandatory data produces an Incomplete decision. Rule-service failure produces no eligibility approval.
Postconditions	The decision and evaluation time are recorded.

UC-02 Record Donation and Units

Field	Specification
Primary actor	Blood Bank Staff
Preconditions	The donor is checked in and approved for donation.

Field	Specification
Trigger	Staff record a completed collection.
Main flow	The system validates collection data, creates one donation identifier, creates unique component unit identifiers and sets each unit to Collected.
Alternate and exception flow	Duplicate identifiers or invalid volume values are rejected.
Postconditions	Donation and unit records exist with an audit event.

UC-03 Record Screening Results

Field	Specification
Primary actor	Blood Bank Staff
Preconditions	A unit exists in Testing state.
Trigger	Staff submit laboratory results.
Main flow	The system validates the test panel. An acceptable complete panel permits release to Available. A reactive, indeterminate or incomplete panel produces Quarantined status.
Alternate and exception flow	Unauthorized access, missing test type or invalid result value is rejected.
Postconditions	The unit status and audit record are updated.

UC-04 Submit Blood Request

Field	Specification
Primary actor	Hospital Staff
Preconditions	The hospital account is active.
Trigger	Hospital staff submit required blood details.
Main flow	The system validates group, component, quantity, priority, required-by time and clinical reference, then creates a Pending request.
Alternate and exception flow	Missing data is rejected. Duplicate client submission returns the existing request when the idempotency key matches.
Postconditions	A traceable Pending request exists.

UC-05 Reserve and Issue Blood Units

Field	Specification
Primary actor	Blood Bank Staff
Preconditions	An Approved request and compatible Available units exist.
Trigger	Staff select units for fulfilment.

Field	Specification
Main flow	The system atomically reserves units, records cross-match results and issues acceptable units. The request becomes Partially Fulfilled or Fulfilled.
Alternate and exception flow	Failed cross-match releases the unit according to policy. Concurrent reservation conflict rejects the later request.
Postconditions	Unit and request states agree and an audit event exists.

UC-06 Manage Accounts and Roles

Field	Specification
Primary actor	System Administrator
Preconditions	The administrator is authenticated.
Trigger	An account is created, changed or deactivated.
Main flow	The system validates the role, branch scope and account status before saving the change.
Alternate and exception flow	The final active administrator cannot be deactivated through the interface.
Postconditions	The account change and audit event are stored.

8 Requirements Traceability

Table 5 Requirements traceability matrix

Requirement	Design allocation	Verification
FR-01	Donor Service	Inspection and TC-F01
FR-02	Donor Service	TC-F01
FR-03	Donor Service	TC-F02
FR-04	Donor Service	TC-F02
FR-05	Donor Service	TC-F03
FR-06	Donor Service	TC-F03
FR-07	Donor Service	TC-F04
FR-08	Donor Service	TC-F05
FR-09	Donation and Screening Service	TC-F06
FR-10	Donation and Screening Service	TC-F06
FR-11	Donation and Screening Service	TC-F07
FR-12	Donation and Screening Service	TC-F07
FR-13	Donation and Screening Service	TC-F07
FR-14	Donation and Screening Service	TC-F08

Requirement	Design allocation	Verification
FR-15	Donation and Screening Service	TC-F08
FR-16	Donation and Screening Service	TC-F08
FR-17	Inventory and Request Service	TC-F09
FR-18	Inventory and Request Service	TC-F10
FR-19	Inventory and Request Service	TC-F11
FR-20	Inventory and Request Service	TC-F11
FR-21	Inventory and Request Service	TC-F12
FR-22	Inventory and Request Service	TC-F13
FR-23	Inventory and Request Service	TC-F13
FR-24	Inventory and Request Service	TC-F13
FR-25	Authentication, Administration or Reporting	TC-S01
FR-26	Authentication, Administration or Reporting	TC-S02
FR-27	Authentication, Administration or Reporting	TC-F14
FR-28	Authentication, Administration or Reporting	TC-F15
FR-29	Authentication, Administration or Reporting	TC-F16
FR-30	Authentication, Administration or Reporting	TC-S03
NFR-01	Cross-cutting architecture	TC-N01
NFR-02	Cross-cutting architecture	TC-N02
NFR-03	Cross-cutting architecture	Operational review
NFR-04	Cross-cutting architecture	TC-N03
NFR-05	Cross-cutting architecture	TC-N04
NFR-06	Cross-cutting architecture	TC-N05
NFR-07	Cross-cutting architecture	TC-N06
NFR-08	Cross-cutting architecture	Code review
NFR-09	Cross-cutting architecture	TC-S02
NFR-10	Cross-cutting architecture	TC-N07
NFR-11	Cross-cutting architecture	TC-S04
NFR-12	Cross-cutting architecture	Inspection
SEC-01	Security architecture	TC-S02
SEC-02	Security architecture	Code review
SEC-03	Security architecture	TC-S01
SEC-04	Security architecture	TC-S05
SEC-05	Security architecture	Configuration inspection

Requirement	Design allocation	Verification
SEC-06	Security architecture	TC-S06
SEC-07	Security architecture	TC-S03
SEC-08	Security architecture	TC-N03
SEC-09	Security architecture	TC-S04
SEC-10	Security architecture	TC-S01

References

[1] ISO IEC IEEE 29148 2018, Systems and software engineering, Life cycle processes, Requirements engineering.

[2] IEEE 829 2008, Standard for Software and System Test Documentation, used as the course-aligned document structure. The standard has been superseded by the ISO IEC IEEE 29119 series.

[3] IEEE 1016 2009, Standard for Information Technology, Systems Design, Software Design Descriptions.

[4] Ministry of Health and Family Welfare, Government of India, National Standards for Blood Centres, Annexure 4 Criteria for Selection of Blood Donor.

[5] Central Drugs Standard Control Organization, Drugs and Cosmetics Act 1940 and Rules 1945, Schedule F provisions for blood banks.

[6] eRaktKosh, Ministry of Health and Family Welfare, overview of centralized donor, testing, inventory and notification functions.

[7] Django Software Foundation, Django documentation for authentication, security and database transactions.

Appendix A Compliance Checklist

Deliverable requirement	Location in this document	Status
SRS in IEEE-aligned format	Sections 1 to 8	Complete
Introduction and descriptive SRS sections	Sections 1 to 3	Complete
Clear functional and non-functional requirements	Section 4	Complete
At least one UML use-case diagram	Section 7, Figure 1	Complete
At least two security objectives and requirements	Section 6	Complete
Traceability with test plan	Section 8	Complete